Shivanshu Srivastava

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Summary

Backend & Systems Engineer and B.Tech CSE student (2026, GPA: 8.49) with hands-on experience building high-performance asynchronous microservices (FastAPI, Motor/MongoDB), applied RAG architectures (ChromaDB, LangChain), and distributed automation systems. Proven track record in national hackathons (SunHacks 2026 Top 10, AICTE Winner) with strong foundations in API architecture, database query optimization, and background event pipelines.

Education

Sandip University **2022 – 2026**
B.Tech, Computer Science and Engineering (GPA: 8.49) *Nashik, Maharashtra*

GlenHill School **2018 – 2021**
SSC & HSC *Varanasi, Uttar Pradesh*

Technical Skills

- **Languages:** Python (AsyncIO), SQL, TypeScript, HTML/CSS
- **Backend & AI:** FastAPI, Motor, REST API Architecture, ChromaDB, LangChain, RAG Pipelines
- **Databases:** MongoDB (Aggregation Pipelines), SQLite, MySQL, SQLAlchemy ORM
- **Systems & Tools:** Wireless ADB, Docker, Tailscale / WireGuard, Linux, Git, GitHub Actions, Postman, JWT

Experience

Brain O Vision **Jan 2024 – Mar 2024**
Machine Learning Intern *Hyderabad, India*

– Preprocessed structured datasets using Pandas and NumPy – handled missing values, encoded categorical features, and applied feature scaling.

– Trained and evaluated classification models (Decision Tree, Random Forest) with Scikit-Learn; measured performance using accuracy, precision, and confusion matrices.

Code Alpha **Nov 2025 – Nov 2025**
Python Programming Intern *Lucknow, India*

– Built modular Python scripts for data processing and file management using standard libraries (os, json, csv, datetime).

– Developed CLI utilities with structured error handling and integrated REST API data fetching using the Requests library.

Projects

TradeFlow – Trading Journal & Analytics Terminal
Live | Python, FastAPI, Motor, MongoDB, Next.js 15, JWT

– Built an asynchronous REST API with FastAPI and Motor (MongoDB) for logging, filtering, and analyzing trades with per-user data isolation.

– Engineered MongoDB aggregation pipelines to calculate portfolio metrics (win rate, net P&L, drawdown limits) and implemented JWT authentication with bcrypt.

ThirdEye – Organizational Memory & RAG System
Source | Python, FastAPI, ChromaDB, LangChain, Node Graphs

- Architected an organizational memory tool with ChromaDB vector search and RAG for conversational document Q&A.
- Built interactive knowledge-graph node visualizations, structured directory management, and user access control; placed in the Top 10 at SunHacks 2026.

ShadowGram – Distributed Publishing Node

Source | Python, Wireless ADB, Tailscale, Docker, Telegram API

- Architected a headless publishing system over a secure Tailscale WireGuard tunnel, automating physical Android device execution via wireless ADB.
- Designed a zero-device-storage staging queue hosted on Telegram Cloud with Docker runner deployment, bypassing Meta Graph API restrictions with automated UI navigation.

NSE Swing Screener

Live | Python, FastAPI, SQLite, BeautifulSoup

- Developed a stock screening service that scrapes daily NSE data and flags potential swing setups based on technical indicators.
- Used SQLite to cache historical data locally and scheduled background scraping jobs with APScheduler to keep results current.

Hackathons

ThirdEye – Org Memory System (Top 10) — April 2026

Sunhacks 2026

- Built the AI chatbot and knowledge-graph visualization features for an organizational memory tool; placed in the Top 10.

HVAC Smart Control System (Top 10) — April 2024

Sunhacks 2024 by Sandip University & ESDS

- Contributed to a predictive ML pipeline for energy-efficient HVAC climate control; team placed in the Top 10.

Sales Prediction Model (Winner) — Oct 2023

AICTE Data Science Hackathon

- Won 1st place by building regression models for sales forecasting and product performance prediction using Scikit-Learn and Pandas.